BIMBINGAN TUGAS AKHIR #8

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1 Deep Neural Network

Diberikan dataset $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^{N}$ dengan N data sampel. Setiap data ke-i dinyatakan sebagai vektor baris $\mathbf{x}_i \in \mathbb{R}^{1\times D}$ dengan D fitur:

$$\mathbf{x}_i = \begin{bmatrix} x_{i,1} & x_{i,2} & \dots & x_{i,D} \end{bmatrix} \in \mathbb{R}^{1\times D} \tag{1}$$

Untuk arsitektur dengan L layer ($\ell = 1, \dots, L$) dan neuron $n_\ell \in \mathbb{N}^+$:

$$\mathbf{x}_i^{(0)} = \mathbf{x}_i \in \mathbb{R}^{1\times n_0}, \quad n_0 = D, \quad x_{i,k}^{(0)} = x_{i,k} \tag{2}$$

di mana $\mathbf{x}_i^{(\ell-1)} \in \mathbb{R}^{1\times n_{\ell-1}}$ ditransformasikan menjadi $\mathbf{x}_i^{(\ell)} \in \mathbb{R}^{1\times n_\ell}$ hingga mencapai output layer $\mathbf{x}_i^{(L)} \in \mathbb{R}^{1\times n_L}$.

1.1 Tinjau Satu Neuron

Untuk neuron ke-j pada layer ℓ ($j = 1, \dots, n_\ell$), komputasi terhadap data ke-i dirumuskan sebagai:

$$z_{i,j}^{(\ell)} = \sum_{k=1}^{n_{\ell-1}} x_{i,k}^{(\ell-1)} w_{kj}^{(\ell)} + b_j^{(\ell)} \tag{3}$$

$$x_{i,j}^{(\ell)} = \sigma^{(\ell)}\left(z_{i,j}^{(\ell)}\right) \tag{4}$$

dengan indeks dan parameter:

- $i \in \{1, \dots, N\}$: indeks data ke-i.
- $k \in \{1, \dots, n_{\ell-1}\}$: indeks fitur input (layer $\ell - 1$).
- $j \in \{1, \dots, n_\ell\}$: indeks neuron / fitur output (layer ℓ).
- $w_{kj}^{(\ell)} \in \mathbb{R}$: bobot fitur k ke neuron j.
- $b_j^{(\ell)} \in \mathbb{R}$: bias neuron j.
- $\sigma^{(\ell)} : \mathbb{R} \to \mathbb{R}$: fungsi aktivasi element-wise.

1.2 Tinjau Satu Data

Didefinisikan vektor baris data fitur ke-i, matriks bobot $W^{(\ell)}$, dan vektor bias untuk layer ℓ:

$$\mathbf{x}_i^{(\ell-1)} = \begin{bmatrix} x_{i,1}^{(\ell-1)} & x_{i,2}^{(\ell-1)} & \dots & x_{i,n_{\ell-1}}^{(\ell-1)} \end{bmatrix} \in \mathbb{R}^{1\times n_{\ell-1}} \tag{5}$$

$$W^{(\ell)} = \begin{bmatrix} w_{11}^{(\ell)} & w_{12}^{(\ell)} & \dots & w_{1n_\ell}^{(\ell)} \\ w_{21}^{(\ell)} & w_{22}^{(\ell)} & \dots & w_{2n_\ell}^{(\ell)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n_{\ell-1}1}^{(\ell)} & w_{n_{\ell-1}2}^{(\ell)} & \dots & w_{n_{\ell-1}n_\ell}^{(\ell)} \end{bmatrix} \in \mathbb{R}^{n_{\ell-1}\times n_\ell} \tag{6}$$

$$\mathbf{b}^{(\ell)} = \begin{bmatrix} b_1^{(\ell)} & \dots & b_{n_\ell}^{(\ell)} \end{bmatrix} \in \mathbb{R}^{1\times n_\ell}, \quad \mathbf{z}_i^{(\ell)} = \begin{bmatrix} z_{i,1}^{(\ell)} & \dots & z_{i,n_\ell}^{(\ell)} \end{bmatrix} \in \mathbb{R}^{1\times n_\ell} \tag{7}$$

Transformasi vektor baris pada layer ℓ untuk data ke-i:

$$\underbrace{\mathbf{z}_i^{(\ell)}}_{(1\times n_\ell)} = \underbrace{\mathbf{x}_i^{(\ell-1)}}_{(1\times n_{\ell-1})} \times \underbrace{W^{(\ell)}}_{(n_{\ell-1}\times n_\ell)} + \underbrace{\mathbf{b}^{(\ell)}}_{(1\times n_\ell)} \tag{8}$$

$$\underbrace{\mathbf{x}_i^{(\ell)}}_{(1\times n_\ell)} = \sigma^{(\ell)}\left(\mathbf{z}_i^{(\ell)}\right) \tag{9}$$

1.3 Tinjau N Data

Diberikan matriks data input $X^{(\ell-1)} \in \mathbb{R}^{N \times n_{\ell-1}}$ di mana setiap baris ke-i memuat vektor baris fitur dari data ke-i, $\mathbf{x}_i^{(\ell-1)}$:

$$X^{(\ell-1)} = \begin{bmatrix} \mathbf{x}_1^{(\ell-1)} \\ \mathbf{x}_2^{(\ell-1)} \\ \vdots \\ \mathbf{x}_i^{(\ell-1)} \\ \vdots \\ \mathbf{x}_N^{(\ell-1)} \end{bmatrix} = \begin{bmatrix} x_{1,1}^{(\ell-1)} & x_{1,2}^{(\ell-1)} & \dots & x_{1,n_{\ell-1}}^{(\ell-1)} \\ x_{2,1}^{(\ell-1)} & x_{2,2}^{(\ell-1)} & \dots & x_{2,n_{\ell-1}}^{(\ell-1)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i,1}^{(\ell-1)} & x_{i,2}^{(\ell-1)} & \dots & x_{i,n_{\ell-1}}^{(\ell-1)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1}^{(\ell-1)} & x_{N,2}^{(\ell-1)} & \dots & x_{N,n_{\ell-1}}^{(\ell-1)} \end{bmatrix} \in \mathbb{R}^{N \times n_{\ell-1}} \tag{10}$$

dengan $X^{(0)} = X \in \mathbb{R}^{N \times D}$ sebagai matriks data awal (N baris data, masing-masing memiliki D kolom fitur).

Operasi tingkat *batch*:

$$\underbrace{Z^{(\ell)}}_{(N \times n_\ell)} = \underbrace{X^{(\ell-1)}}_{(N \times n_{\ell-1})} \times \underbrace{W^{(\ell)}}_{(n_{\ell-1} \times n_\ell)} + \underbrace{\mathbf{1}_N}_{(N \times 1)} \times \underbrace{\mathbf{b}^{(\ell)}}_{(1 \times n_\ell)} \tag{11}$$

$$\underbrace{X^{(\ell)}}_{(N \times n_\ell)} = \sigma^{(\ell)}\left(Z^{(\ell)}\right) \tag{12}$$

dengan $\mathbf{1}_N = [1, 1, \dots, 1]^T \in \mathbb{R}^{N \times 1}$ adalah vektor kolom bernilai satu. Baris ke-i dari matriks $Z^{(\ell)}$ dan $X^{(\ell)}$ masing-masing menyatakan $\mathbf{z}_i^{(\ell)} \in \mathbb{R}^{1 \times n_\ell}$ dan $\mathbf{x}_i^{(\ell)} \in \mathbb{R}^{1 \times n_\ell}$ dari data ke-i.

1.4 Fungsi Objektif

Pada layer terakhir ($\ell = L$), tidak diterapkan fungsi aktivasi non-linier, sehingga prediksi output berpadanan langsung dengan transformasi matriks:

$$\hat{Y} = X^{(L-1)} W^{(L)} + \mathbf{1}_N \mathbf{b}^{(L)} \in \mathbb{R}^{N \times n_L} \tag{13}$$

dan untuk data tunggal ke-i:

$$\hat{\mathbf{y}}_i = \mathbf{x}_i^{(L-1)} W^{(L)} + \mathbf{b}^{(L)} \in \mathbb{R}^{1 \times n_L} \tag{14}$$

Diberikan matriks target $Y \in \mathbb{R}^{N \times n_L}$. Fungsi objektif *Mean Squared Error* (MSE) dievaluasi melalui selisih target Y dengan prediksi $\hat{Y}$:

$$\mathcal{L} = \frac{1}{2N} \left\| Y - \hat{Y} \right\|_F^2 = \frac{1}{2N} \sum_{i=1}^{N} \sum_{j=1}^{n_L} (y_{i,j} - \hat{y}_{i,j})^2 \tag{15}$$

di mana $\| \cdot \|_F$ menyatakan norma Frobenius matriks.

2 Backward Propagation

2.1 Motivasi

Pembaruan parameter matriks bobot $W^{(\ell)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell}$ dan vektor bias $\mathbf{b}^{(\ell)} \in \mathbb{R}^{1 \times n_\ell}$ pada algoritma *Gradient Descent* dengan *learning rate* $\eta > 0$ dirumuskan sebagai:

$$W_{n+1}^{(\ell)} = W_n^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial W_n^{(\ell)}} \tag{16}$$

$$\mathbf{b}_{n+1}^{(\ell)} = \mathbf{b}_n^{(\ell)} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{b}_n^{(\ell)}} \tag{17}$$

dengan n menyatakan indeks iterasi.

2.2 Turunan Parameter pada Layer Output ($\ell = L$)

Karena pada layer output tidak diterapkan fungsi aktivasi, prediksi dinyatakan sebagai:

$$\hat{Y} = X^{(L-1)} W^{(L)} + \mathbf{1}_N \mathbf{b}^{(L)} \in \mathbb{R}^{N \times n_L} \tag{18}$$

Claim 2.1. *Turunan fungsi objektif $\mathcal{L}$ terhadap matriks prediksi $\hat{Y} \in \mathbb{R}^{N \times n_L}$ adalah:*

$$\frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N}(\hat{Y} - Y) \in \mathbb{R}^{N \times n_L} \tag{19}$$

Proof. Turunan fungsi skalar $\mathcal{L}$ terhadap matriks $\hat{Y}$ didefinisikan sebagai:

$$\frac{\partial \mathcal{L}}{\partial \hat{Y}} = \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}_1} \\ \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}_2} \\ \vdots \\ \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}_i} \\ \vdots \\ \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}_N} \end{bmatrix} = \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial \hat{y}_{1,1}} & \frac{\partial \mathcal{L}}{\partial \hat{y}_{1,2}} & \cdots & \frac{\partial \mathcal{L}}{\partial \hat{y}_{1,n_L}} \\ \frac{\partial \mathcal{L}}{\partial \hat{y}_{2,1}} & \frac{\partial \mathcal{L}}{\partial \hat{y}_{2,2}} & \cdots & \frac{\partial \mathcal{L}}{\partial \hat{y}_{2,n_L}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \mathcal{L}}{\partial \hat{y}_{i,1}} & \frac{\partial \mathcal{L}}{\partial \hat{y}_{i,2}} & \cdots & \frac{\partial \mathcal{L}}{\partial \hat{y}_{i,n_L}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \mathcal{L}}{\partial \hat{y}_{N,1}} & \frac{\partial \mathcal{L}}{\partial \hat{y}_{N,2}} & \cdots & \frac{\partial \mathcal{L}}{\partial \hat{y}_{N,n_L}} \end{bmatrix} \in \mathbb{R}^{N \times n_L} \tag{20}$$

Dengan *chain rule*, turunan terhadap setiap elemen $\hat{y}_{i,j}$:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}_{i,j}} = \frac{\partial \mathcal{L}}{\partial (y_{i,j} - \hat{y}_{i,j})} \frac{\partial (y_{i,j} - \hat{y}_{i,j})}{\partial \hat{y}_{i,j}} = \frac{1}{N}(y_{i,j} - \hat{y}_{i,j}) \cdot (-1) = \frac{1}{N}(\hat{y}_{i,j} - y_{i,j}) \tag{21}$$

Sehingga diperoleh $\frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N}(\hat{Y} - Y)$. □

Claim 2.2. *Gradien fungsi objektif terhadap matriks bobot $W^{(L)}$ adalah:*

$$\frac{\partial \mathcal{L}}{\partial W^{(L)}} = \frac{1}{N} \left(X^{(L-1)} \right)^T \left(\hat{Y} - Y \right) \in \mathbb{R}^{n_{L-1} \times n_L} \tag{22}$$

Proof. Dengan *chain rule* terhadap $\hat{Y} = X^{(L-1)} W^{(L)} + \mathbf{1}_N \mathbf{b}^{(L)}$:

$$\frac{\partial \mathcal{L}}{\partial W^{(L)}} = \frac{\partial \mathcal{L}}{\partial \hat{Y}} \frac{\partial \hat{Y}}{\partial W^{(L)}} = \left(X^{(L-1)} \right)^T \frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N} \left(X^{(L-1)} \right)^T \left(\hat{Y} - Y \right) \tag{23}$$

□

Claim 2.3. *Gradien fungsi objektif terhadap vektor bias $\mathbf{b}^{(L)}$ adalah:*

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L)}} = \frac{1}{N} \mathbf{1}_N^T \left(\hat{Y} - Y \right) \in \mathbb{R}^{1 \times n_L} \tag{24}$$

Proof. Dengan *chain rule* terhadap $\hat{Y} = X^{(L-1)} W^{(L)} + \mathbf{1}_N \mathbf{b}^{(L)}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L)}} = \frac{\partial \mathcal{L}}{\partial \hat{Y}} \frac{\partial \hat{Y}}{\partial \mathbf{b}^{(L)}} = \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \hat{Y}} = \frac{1}{N} \mathbf{1}_N^T \left(\hat{Y} - Y \right) \tag{25}$$

□

2.3 Turunan Parameter pada Layer $\ell = L-1, \ldots, 2, 1$

Pada hidden layer $\ell \in \{L-1, \ldots, 2, 1\}$, fungsi aktivasi $\sigma^{(\ell)}$ diterapkan terhadap bentuk matriks:

$$X^{(\ell)} = \sigma^{(\ell)}\left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right) \in \mathbb{R}^{N \times n_\ell} \tag{26}$$

***Backward Propagation* pada Layer $\ell = L-1$:**

Backward propagation galat dari layer output ke fitur $X^{(L-1)}$ dievaluasi melalui aturan rantai:

$$\frac{\partial \mathcal{L}}{\partial X^{(L-1)}} = \frac{\partial \mathcal{L}}{\partial \hat{Y}} \frac{\partial \hat{Y}}{\partial X^{(L-1)}} = \frac{1}{N}\left(\hat{Y} - Y\right)\left(W^{(L)}\right)^T \in \mathbb{R}^{N \times n_{L-1}} \tag{27}$$

Turunan terhadap bentuk pre-aktivasi layer $L-1$:

$$\frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} = \frac{\partial \mathcal{L}}{\partial X^{(L-1)}} \odot \sigma^{(L-1)\prime}\left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right) \tag{28}$$

Gradien parameter $W^{(L-1)}$ dan $\mathbf{b}^{(L-1)}$:

$$\frac{\partial \mathcal{L}}{\partial W^{(L-1)}} = \left(X^{(L-2)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \in \mathbb{R}^{n_{L-2} \times n_{L-1}} \tag{29}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L-1)}} = \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \in \mathbb{R}^{1 \times n_{L-1}} \tag{30}$$

Backward propagation ke fitur layer $L-2$:

$$\frac{\partial \mathcal{L}}{\partial X^{(L-2)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(L-2)}W^{(L-1)} + \mathbf{1}_N \mathbf{b}^{(L-1)}\right)} \left(W^{(L-1)}\right)^T \in \mathbb{R}^{N \times n_{L-2}} \tag{31}$$

***Backward Propagation* pada Layer $\ell = L-2$:**

Turunan terhadap bentuk pre-aktivasi layer $L-2$:

$$\frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} = \frac{\partial \mathcal{L}}{\partial X^{(L-2)}} \odot \sigma^{(L-2)\prime}\left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right) \tag{32}$$

Gradien parameter $W^{(L-2)}$ dan $\mathbf{b}^{(L-2)}$:

$$\frac{\partial \mathcal{L}}{\partial W^{(L-2)}} = \left(X^{(L-3)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \in \mathbb{R}^{n_{L-3} \times n_{L-2}} \tag{33}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(L-2)}} = \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \in \mathbb{R}^{1 \times n_{L-2}} \tag{34}$$

Backward propagation ke fitur layer $L-3$:

$$\frac{\partial \mathcal{L}}{\partial X^{(L-3)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(L-3)}W^{(L-2)} + \mathbf{1}_N \mathbf{b}^{(L-2)}\right)} \left(W^{(L-2)}\right)^T \in \mathbb{R}^{N \times n_{L-3}} \tag{35}$$

***Backward Propagation* Umum pada Hidden Layer ($\ell = L-1, \ldots, 2, 1$):**

Secara umum, untuk setiap hidden layer $\ell \in \{L-1, \ldots, 2, 1\}$:

$$\frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} = \frac{\partial \mathcal{L}}{\partial X^{(\ell)}} \odot \sigma^{(\ell)\prime}\left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right) \in \mathbb{R}^{N \times n_\ell} \tag{36}$$

$$\frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \left(X^{(\ell-1)}\right)^T \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \in \mathbb{R}^{n_{\ell-1} \times n_\ell} \tag{37}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(\ell)}} = \mathbf{1}_N^T \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \in \mathbb{R}^{1 \times n_\ell} \tag{38}$$

$$\frac{\partial \mathcal{L}}{\partial X^{(\ell-1)}} = \frac{\partial \mathcal{L}}{\partial \left(X^{(\ell-1)}W^{(\ell)} + \mathbf{1}_N \mathbf{b}^{(\ell)}\right)} \left(W^{(\ell)}\right)^T \in \mathbb{R}^{N \times n_{\ell-1}} \tag{39}$$

dengan kondisi batas input awal $X^{(0)} = X \in \mathbb{R}^{N \times D}$.